Grade-Aware Dual-Level Difficulty Control for Multi-Agent Vietnamese Text Summarization in Primary Education

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***Abstract*—Automatic summarization can help primary-school students identify main ideas, yet most systems emit a single output style and do not control linguistic difficulty by grade. Building on our previous Vietnamese educational summarizer and multi-agent prototype, this paper presents grade-aware dual-level difficulty control for Vietnamese text summarization in primary education (Grades 1–5). The first level selects a target grade through cumulative grade-stratified vocabularies. The second level sets an Easy, Medium, or Hard mode within that grade using a readability score. The controller is shared by a multi-agent workflow that plans the task, drafts an extractive or abstractive summary with fine-tuned PhoBERT and ViT5 models, evaluates the draft, and revises it when grade–mode constraints are not met. On an educational corpus of more than 10,000 texts and 22,000 summary pairs, the abstractive agent attains a ROUGE-1 of 0.82 and a BERTScore of 0.76. A human evaluation with primary-school teachers indicates that the summaries are readable and suitable for classroom use, with remaining gaps in preserving the educational message.**

Keywords—Text summarization, Multi-agent systems, Grade-aware readability, Vietnamese NLP, Educational AI, Primary education

I. INTRODUCTION

Summarization is used in primary education as a pedagogical practice that can support reading comprehension [1, 2]. Automatic systems can isolate main ideas and reduce redundant content, but they usually emit a single output style and offer little control over length, vocabulary, or syntactic complexity for a stated grade [3, 4].

The issue is particularly relevant to Vietnamese primary education. Word segmentation, compound words, and Sino-Vietnamese items make it easy for unconstrained rewriting to exceed the vocabulary of Grades 1–5. In prior work, we built an application that fine-tunes neural models to summarize Vietnamese readings and to constrain length and vocabulary by grade [5], and a multi-agent prototype that assigns extractive and abstractive generation to specialized agents with grade-level lexicons [6]. Both systems adapt output to a target grade, yet they do not separate which grade from how difficult the text should be within that grade, and they do not treat a missed readability target as a condition for revision.

Multi-agent summarization can coordinate planning, generation, and critique [6–8]. For young learners, that coordination has limited educational value unless the agents share an explicit difficulty objective. This paper therefore formalizes grade-aware dual-level difficulty control. Level 1 is the target grade (1–5), encoded by cumulative vocabularies. Level 2 is an Easy, Medium, or Hard mode inside that grade, obtained by mapping a readability score to grade-specific thresholds. The controller is shared by planning, draft generation with fine-tuned PhoBERT and ViT5, and evaluation-driven revision. Extending [5] and [6], this paper makes three contributions:

(1) A dual-level controller that jointly enforces a primary grade and an Easy/Medium/Hard mode, using grade-stratified dictionaries and a readability score that prioritizes vocabulary suitability.

(2) An integration of the controller into the multi-agent workflow so that planning, extractive/abstractive generation, and revision operate under the same grade–mode constraints.

(3) An evaluation on an expanded educational corpus (more than 10,000 texts and 22,000 summary pairs), combining ROUGE and BERTScore with readability measures and a human evaluation with primary-school teachers.

The remainder of the paper is organized as follows. Section II reviews related work. Section III presents the controller and its integration. Section IV reports experiments. Section V concludes.

II. RELATED WORK

Surveys of automatic text summarization organize the field into extractive selection and abstractive generation, and they document evaluation practice based mainly on ROUGE-style overlap and embedding similarity [3, 4]. Luo et al. treat controllability as an emerging axis—length, style, or content focus—but the reviewed systems are still benchmarked on news and scientific text [3]. Supriyono et al. similarly list evaluation and domain transfer among open challenges, without a protocol that binds a school grade to an in-grade difficulty band [4]. The literature therefore explains how to rank sentences or rewrite them; it does not specify when a draft is educationally unacceptable.

Work on summarization in primary classrooms addresses a different object. Ramirez-Avila and Barreiro study student-authored summaries of narrative texts as a comprehension exercise [1]. Avivah et al. report that guided summary practice can improve reading skills in elementary settings [2]. Both results support the claim that wording must stay within a child's reach. They do not, however, describe an automatic generator, a grade-stratified lexicon, or a numeric readability target that a system could test before releasing an output.

Multi-agent summarization moves from a single decoder to coordinated roles. Durak et al. instantiate agents for educational summarization and compare the outputs with human writing, with emphasis on quality relative to teachers or students rather than on grade-indexed vocabulary ceilings [7]. Chen et al. add an auto-evaluation agent in the medical domain, so revision is triggered by semantic or factual checks, not by a primary-grade readability score [8]. Our multi-agent prototype for Vietnamese readings already splits extractive PhoBERT [9] and abstractive ViT5 [10] generation and attaches grade lexicons [6]. Relative to [7] and [8], [6] changes the language and the learner group; relative to the present paper, it still uses the grade label as a lexical resource rather than as one half of a two-level control objective.

Vietnamese pretrained models supply the encoders and decoders used above. PhoBERT is trained for general Vietnamese understanding [9]; ViT5 is trained for Vietnamese text-to-text generation [10]. The VLSP abstractive multi-document collection enlarges the pool of Vietnamese summary pairs, but its documents are not labeled by primary grade [11]. Fine-tuning these backbones on textbook and children's readings [5] reduces the domain mismatch; it does not by itself define Easy, Medium, and Hard operating points inside each grade, nor an acceptance test tied to those points. Section III turns that missing control interface into an explicit dual-level mechanism.

III. Proposed Method

A. System Overview

We present MAS-ViSum as a multi-agent pipeline for Vietnamese summarization in primary education. As shown in Fig. 1, a request is first classified by intent: a new summary, an adjustment of an existing text, or a supporting operation such as clarification or recognition of text from an image. When the input is incomplete or is supplied as an image, that supporting step is completed before summarization begins. The summarization path is constrained at two levels: a target grade (1–5) and a difficulty mode (Easy, Medium, or Hard). A planner uses these constraints to choose extractive or abstractive generation and to set the acceptance criteria for the draft. The generated summary is then checked for vocabulary consistent with the target grade and for semantic similarity to the source; if either check fails, the planner issues another draft, up to a fixed number of iterations. The same grade and mode are applied when the intent is to simplify or enrich a text that has already been produced. In both cases the two constraints act during planning and during acceptance of the output.

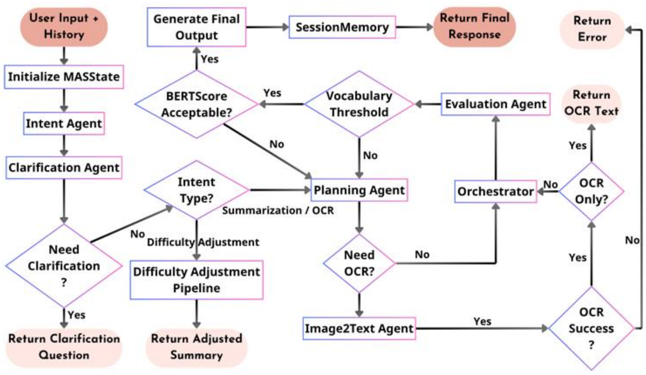

Fig. 1. Overall architecture of the proposed MAS-ViSum framework. The system integrates multiple specialized agents coordinated by an Orchestrator, with dynamic grade-aware difficulty control and iterative refinement.

B. Multi-Agent System

The core of MAS-ViSum is a LangGraph-based multi-agent system that coordinates multiple specialized agents to achieve adaptive and high-quality Vietnamese text summarization. The planner records the target grade and the difficulty mode and selects extractive or abstractive generation. The extractor [9] or the abstracter [10] then produces a draft, and the evaluation agent tests that draft against the same grade–mode pair. The orchestrator advances this sequence and stops at acceptance or at a fixed iteration limit. This architecture decomposes the complex summarization task into distinct responsibilities, enabling greater modularity, collaboration, and iterative refinement. Table 1 summarizes the main agents and their corresponding functions in the proposed framework.

Table. 1. Agent Descriptions in the MAS-ViSum Framework

Agent	Function
Intent Agent	Classifies user intent and extracts key requirements
Image2Text Agent	Performs OCR to extract text from input images
Clarification Agent	Generates clarification questions when input is ambiguous
Planning Agent	Creates an execution plan based on intent, grade level, and difficulty mode
Extractor Agent	Performs extractive summarization using fine-tuned PhoBERT
Abstracter Agent	Generates abstractive summaries using fine-tuned ViT5
Evaluation Agent	Assesses summary quality using semantic and readability metrics
Memory Agent	Manages short-term session memory and long-term experience via ChromaDB

C. Dual-level Difficulty Control

1) Motivation and Problem Definition

Traditional text summarization systems typically focus on surface-level control such as length or compression ratio, while paying limited attention to linguistic difficulty and pedagogical suitability. This limitation is particularly critical in primary education, where summaries must match students' reading ability, vocabulary knowledge, and cognitive development. In Vietnamese, additional challenges arise from sentence structure flexibility, tonal marks, and frequent use of Sino-Vietnamese words. To address these issues, we propose a dual-level difficulty control mechanism that dynamically adjusts summary complexity according to both grade level and user-specified difficulty mode.

- Grade level (Grades 1 to 5), aligned with the official Vietnamese primary school curriculum.
- Difficulty mode (Easy, Medium, Hard) within each grade.

This representation allows flexible personalization while maintaining curriculum compatibility.

2) Linguistic Feature Design

In order to quantify readability, the system extracts a comprehensive set of linguistic features at both lexical and syntactic levels. Key features include vocabulary coverage relative to grade-specific dictionaries, rare and unknown word ratios, average word grade, average sentence length, long sentence ratio, and Sino-Vietnamese word density. These features are computed using the grade-stratified Vietnamese vocabulary dictionaries developed in this study. The overall readability of a summary is measured by a weighted composite score:

$$RS(g, d) = w_1 * V_g + w_2 * (1 - R_r) + w_3 * (1 - S_{norm}) + w_4 * H_v \quad (1)$$

where:

- V_g is the grade-specific vocabulary coverage ratio,
- R_r is the rare/unknown word ratio,
- S_{norm} is the normalized average sentence length,
- H_v is the Sino-Vietnamese word ratio,
- w_1, w_2, w_3, w_4 are the weighting coefficients.

The weights were empirically tuned based on preliminary experiments and feedback from primary school teachers, resulting in $w_1 = 0.40$, $w_2 = 0.25$, $w_3 = 0.20$, and $w_4 = 0.15$. This configuration assigns the highest priority to vocabulary suitability, which is the most critical factor for young learners' comprehension. The final difficulty level (Easy, Medium, or Hard) is determined by mapping the readability score to predefined thresholds for each grade.

3) Control Strategy

The system employs different control strategies for extractive and abstractive summarization. For extractive summarization, difficulty is primarily adjusted by varying the number of selected sentences. For abstractive summarization, a post-generation refinement pipeline is applied. Algorithm 1 as in Fig. 2 details this difficulty-controlled adjustment process, which includes vocabulary simplification or controlled enrichment, sentence shortening, and semantic fidelity preservation. The difficulty controller is tightly integrated with the Evaluation Agent, allowing iterative adjustment until both semantic quality and readability requirements are satisfied.

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Algorithm 1: DIFFICULTY_CONTROLLED_ADJUSTMENT

Input: draftSummary, targetGrade, targetDiff
Output: adjustedSummary

BEGIN
  current ← draftSummary
  feats ← EXTRACT_FEATURES(current)
  score ← COMPUTE_SCORE(feats, targetGrade)
  iter ← 0
  WHILE (NOT MATCH(score, targetDiff)) AND
          (iter < MAX_ITER) DO
     IF score > UPPER_THRESHOLD(targetDiff)
                                          THEN
       current ← SIMPLIFY(current,targetGrade)
     ELSE
       current ← ENRICH(current,targetGrade)
     END IF
      feats ← EXTRACT_FEATURES(current)
      score ← COMPUTE_SCORE(feats,targetGrade)
      iter ← iter + 1
  END WHILE
  RETURN current
END
```

Fig. 2. Pseudo-code of the Difficulty-Controlled Summarization Algorithm

Algorithm 1 returns the draft when its score matches the requested grade–mode interval or when the iteration limit is reached.

D. Summarization models

To support both extractive and abstractive summarization within the multi-agent framework, we fine-tuned two Vietnamese pretrained language models tailored to the primary education domain.

1) Extractive Summarization Model

For extractive summarization, we fine-tuned PhoBERT [9] as a sentence importance classifier. Each sentence, together with its surrounding context, is fed into the model to predict whether it should be included in the summary. The model was trained on more than 10,000 educational documents, yielding over 240,000 labeled sentence samples. Training was optimized using the macro-averaged F1 score, with data partitioned at the document level to avoid leakage. Sentence selection combines graph-based ranking methods with PhoBERT's semantic scoring and the Maximal Marginal Relevance (MMR) criterion to balance relevance and diversity:

$$MMR(s_i) = \lambda.Score(s_i) - (1-\lambda)\max_{s_j \in R}(E(s_i)^{\top}E(s_i)) \quad (2)$$

where *E(.)* denotes the embedding vector produced by PhoBERT, Score(s_i, D) is the relevance score between sentence s_i and the full document D, and R is the set of already selected sentences. The parameter lambda λ controls the trade-off between relevance and redundancy. The final extractive summary is formed by concatenating the selected sentences in their original order.

2) Abstractive Summarization Model

For abstractive summarization, we fine-tuned ViT5 [10] on more than 22,000 high-quality (input, summary) pairs from educational texts. In addition to the standard negative log-likelihood loss, we incorporated ROUGE-L and BERTScore into the training objective to improve semantic fidelity and fluency. Hyper-parameter search was conducted using Optuna. The best configuration used a learning rate of 1.48×10^{-4}, batch size of 8, and 5 epochs. Techniques such as mixed precision training, gradient accumulation, warmup scheduling, and early stopping were employed to enhance training stability and efficiency.

IV. Experiments and Results

A. Dataset and Preprocessing

To train and evaluate the proposed models, we constructed a large-scale educational dataset specifically for Vietnamese primary school students. The corpus was collected from authentic and high-quality sources aligned with the national curriculum, including official textbooks, children's literature, folk tales, and informative texts. The main data sources and their statistics are summarized in Table 2.

Table. 2. Dataset Sources and Distribution

Data Source	Type	Count	Publisher
Canh Dieu Textbooks	Readings, Short stories	652	Vietnam National University of Education Press
Ket Noi Tri Thuc Textbooks	Readings, Short stories	987	Vietnam Education Publishing House
Chan Troi Sang Tao Textbooks	Readings, Short stories	1,267	Vietnam Education Publishing House
Vietnamese Fairy Tales	Folk Tales	1,500	Kim Dong Publishing House
Vietnamese & World Fables	Fables	1,500	Youth Publishing House
Vietnam Children's Literature	Modern Tales	1,500	Kim Dong Publishing House
Vietnamese History	Informative Texts	2,720	Social Sciences Publishing House

To support the difficulty control mechanism, we also constructed grade-stratified Vietnamese vocabulary dictionaries. Using VnCoreNLP for word segmentation and normalization, we built cumulative vocabularies for Grades 1 to 5, where each higher grade inherits all words from previous grades while adding new ones. The vocabulary sizes are shown in Table 3.

Table. 3. Grade-Level Vietnamese Vocabulary dictionary

Grade	Dictionary Name	Vocabulary Size
1	grade_1.txt	3,102
2	grade_2.txt	5,314
3	grade_3.txt	7,689
4	grade_4.txt	10,959
5	grade_5.txt	13,677

For extractive summarization, documents were sentence-segmented and labeled, producing more than 240,000 training samples. Each sentence was then ranked using a unified extractive scoring function that combines graph-based importance, semantic relevance, and redundancy reduction to determine the most informative sentences for the final summary.

$$S^* = arg \max_{R \subseteq D, |R| = k_m} \sum_{s_i \in R} [\alpha C(s_i) + \beta Rel(s_i, D) - \gamma Red(s_i, R)] \quad (3)$$

where S^* is the final extractive summary, R is the selected sentence subset, k_m is the number of selected sentences, $C(s_i)$ is the combined TextRank–LexRank score, $Rel(s_i, D)$ is the semantic relevance between sentence s_i and document D, $Red(s_i, R)$ is the redundancy score against previously selected sentences and α, β, γ are weighting coefficients.

For abstractive summarization, we generated high-quality summary pairs, resulting in over 22,000 samples. This process was formulated as a constrained optimization problem that jointly maximizes grade-level vocabulary compatibility and semantic preservation while controlling summary length.

$$S^* = arg \max_{s_i \in C} [\alpha Coverage(S_i, V_G) + \beta Sem(S_i, D) - \gamma LenPenalty(S_i)] \quad (4)$$

where S^* is final abstractive summary, C is the set of candidate summaries generated by LLM, $Coverage(S_i, V_G)$ vocabulary suitability at grade level, $Sem(S_i, D)$ semantic similarity between summary and document, $LenPenalty(S_i)$ length penalty function for exceeding the desired summary length and α, β, γ are weighting coefficients.

All data splits were performed at the document level (80/10/10 for train/validation/test) to ensure no information leakage between splits. Text preprocessing steps included cleaning, normalization, and removal of noisy content to maintain high data quality.

B. Experimental Setup

All experiments were conducted on GPU-accelerated computing environments. Initial development and hyper-parameter tuning were performed on Kaggle notebooks. The main training runs were executed on a cloud platform (Vast.ai) using NVIDIA RTX 3090 GPUs. Table 4 reports the Optuna-selected learning rate, batch size, and number of epochs for PhoBERT and ViT5. During inference, the system assesses each generated summary based on semantic quality (ROUGE and BERTScore) and readability (vocabulary coverage). If the summary fails to meet the predefined thresholds (e.g., vocabulary coverage ratio below 0.8), the system triggers up to three iterative refinement cycles. For abstractive summarization, three output length modes were supported: short (~30%), medium (~40%), and long (~60%) of the original document length. Evaluation was also conducted on 4,983 VietJack passages that were not used for training or validation. Unlike the held-out split, these passages do not come from the same source collection as the training data.

Table. 4. Hyper-parameters for Fine-tuning Models

Model	Task	Learning Rate	Batch Size	Epochs	Optimization Method
PhoBERT	Extractive	2.24×10^{-5}	16	6	Optuna, F1 Macro
ViT5	Abstractive	1.48×10^{-4}	8	5	Optuna, ROUGE-L, BERTScore

This study conducts a comparative analysis against several widely adopted text summarization methods. Both models are evaluated and compared against Lead-n, TextRank, LexRank, and BertSum, and with Qwen2.5-3B-Instruct, Llama-3.2-3B-Instruct, and GPT-4o mini, with the aim of assessing their capacity for salient content selection, summary generation quality, fluency of expression, and suitability for primary school students. ROUGE-1, ROUGE-2, and ROUGE-L are employed to evaluate the degree of information preservation between the generated summaries and the reference texts. In addition, BERTScore was used to assess semantic similarity based on contextual embeddings from Transformer-based models, which is particularly suitable for evaluating abstractive summarization. Readability was further quantified through a comprehensive set of linguistic features, including total words, unique words, rare word ratio, unknown word ratio, average word grade, number of sentences, average sentence length, long sentence ratio, average word length, words per sentence, and Sino-Vietnamese word count. Five primary-school teachers also rated a sample of outputs on a five-point Likert scale along seven criteria: content fidelity, coverage, readability, grade-level vocabulary, coherence, lesson/message preservation, and overall educational suitability.

C. Results

1) Automatic Evaluation

Based on Table 5, the VAS model achieves a BERTScore of 0.76 and ROUGE-1 of 0.82, higher than Qwen and LLaMA in semantic preservation and key information retention. ROUGE-2 (0.55) and ROUGE-L (0.56) also remain competitive, although they do not surpass GPT. However, ROUGE and BERTScore may not fully reflect the practical effectiveness of VAS, as these metrics primarily measure lexical overlap and semantic similarity without explicitly considering readability or grade-level appropriateness. This limitation is particularly relevant to Vietnamese, where word segmentation, compound words, Sino-Vietnamese vocabulary, and flexible sentence structures can affect lexical and semantic matching. Moreover, VAS is designed with readability-oriented objectives, including grade-level vocabulary control, simpler sentence structures, and reduced use of difficult words. Therefore, automatic metrics should be interpreted together with readability measures and human evaluation to provide a more comprehensive assessment of the model.

VES yields lower scores than the strongest baselines, particularly TextRank and LexRank. One possible factor is the characteristics of the primary-school dataset, especially for Grades 1 and 2, where many texts contain relatively few sentences and limited vocabulary. Such short and linguistically simple passages provide fewer sentence-level candidates and less contextual information for learning-based sentence selection, which may introduce variability in sentence importance estimation. In contrast, graph-based methods can benefit from the concise structure and explicitly expressed main ideas of these texts. Nevertheless, VES maintains stable overall performance and offers greater flexibility for integration with the proposed readability control and multi-agent mechanisms.

Table. 5. Evaluation Results on Baseline Methods

Method / Models	Type Summary	ROUGE-1	ROUGE-2	ROUGE-L	BERT Score
Lead-n	Extractive	0.78	0.53	0.52	0.73
TextRank	Extractive	0.83	0.65	0.66	0.79
LexRank	Extractive	0.82	0.62	0.63	0.78
BertSum	Extractive	0.81	0.59	0.61	0.75
VES	Extractive	0.77	0.52	0.53	0.71
Qwen2.5-3B	Abstractive	0.71	0.45	0.47	0.71
Llama3.2-3B	Abstractive	0.68	0.43	0.45	0.69
GPT-4o mini	Abstractive	0.86	0.64	0.66	0.80
VAS	Abstractive	0.82	0.55	0.56	0.76

2) Readability Evaluation

The results as in Table 6 indicate that all models generate summaries shorter than the reference text, with Qwen and LLaMA producing the lowest word counts but exhibiting higher rare word ratios (0.078) and unknown word ratios (0.067).

Table. 6. Mean readability features per summary for reference texts, three LLM baselines, and VAS

Indicators	References	Qwen 2.5-3B	Llama 3.2-3B	GPT-4o mini	VAS
total_words	152.206	102.828	89.543	136.813	148.028
unique_words	98.591	69.910	59.772	92.552	97.124
rare_word_ratio	0.057	0.078	0.066	0.062	0.060
unknown_word_ratio	0.050	0.067	0.057	0.053	0.052
avg_word_grade	1.158	1.211	1.187	1.182	1.167
num_sentences	9.996	7.374	6.247	7.720	8.698
avg_sentence_length	14.480	14.108	14.534	16.894	16.423
long_sentence_ratio	0.458	0.392	0.402	0.671	0.630
avg_word_length	3.728	3.760	3.769	3.830	3.751
words_per_sentence	15.384	14.874	15.386	17.798	17.335
hanviet_word_count	25.921	17.042	14.671	22.993	25.660

Regarding lexical difficulty, the proposed model achieves the lowest avg_word_grade of 1.167, closely approximating the reference text (1.158). The model further generates an average of 8.70 sentences with 16.42 words per sentence and a long_sentence_ratio of 0.630, while maintaining a Sino-Vietnamese word count closely aligned with the reference text (25.66 versus 25.92). As illustrated in Fig. 3, the grade-level readability results demonstrate a generally consistent progression in the summaries generated by VAS. From Grades 2 to 5, the average summary length increases from 73.45 to 168.23 words, while the number of sentences increases from 4.23 to 9.95, indicating that the system gradually provides more information as the target grade increases. Vocabulary complexity also exhibits a moderate grade-dependent pattern, with the average word grade increasing from 1.154 in Grade 2 to 1.175 in Grade 5, while the number of Sino-Vietnamese words increases from 12.37 to 29.15. In contrast, sentence-level characteristics remain relatively stable, with average sentence length ranging from 16.25 to 16.94 words, suggesting that the current control mechanism differentiates grade levels more strongly through content amount and vocabulary selection than through syntactic complexity. Overall, these results provide evidence that the proposed grade-level control mechanism can produce distinguishable readability profiles across grade levels, although stronger sentence-structure control remains an area for further improvement.

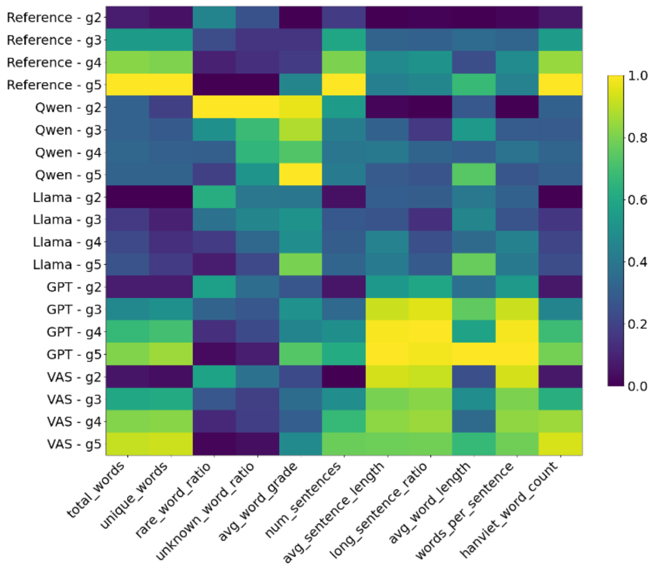

Fig. 3. Baseline Comparison Across Grade Levels Using Grade-Based Readability Control Indicators

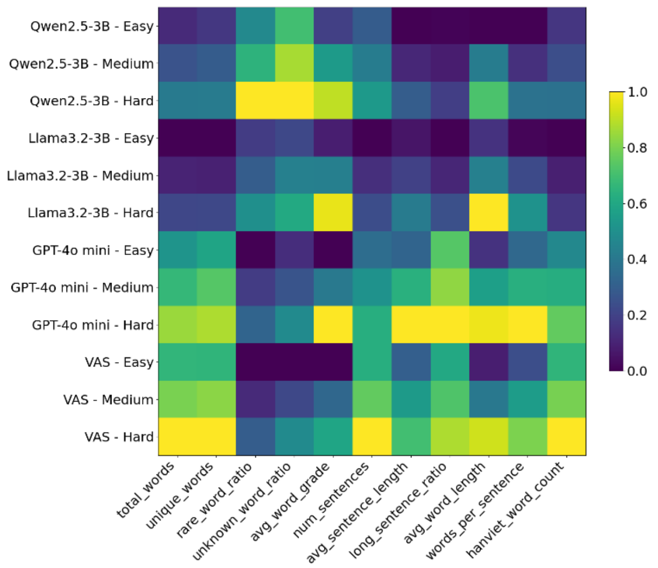

Fig. 4. Comparison of Readability Difficulty Levels (Easy, Medium, and Hard) Based on Reading Difficulty Evaluation Indicators

As illustrated in Fig. 4, the proposed readability-control mechanism exhibits a consistent progression from Easy to Medium and Hard across multiple linguistic dimensions. Summary length increases from 136.334 to 164.813 words, while vocabulary complexity (avg_word_grade) rises from 1.098 to 1.220. Sentence-level complexity follows a similar pattern, with average sentence length increasing from 15.112 to 17.230 words and the long-sentence ratio from 0.586 to 0.686. Other indicators, including vocabulary diversity, rare-word usage, word length, and Sino-Vietnamese vocabulary, generally increase with difficulty. Overall, the heatmap demonstrates distinguishable difficulty profiles across the three levels, indicating that the proposed mechanism controls not only summary length but also lexical and sentence-level characteristics.

3) Teacher Evaluation

Five primary-school teachers rated a sample of summaries on a five-point Likert scale as in Table 7 for content fidelity, key-information coverage, readability, grade-level vocabulary, coherence, lesson/message preservation, and overall educational suitability. Mean scores were 4 for fidelity, coverage, coherence, and overall suitability; 5 for readability and grade-level vocabulary; and 3 for lesson/message preservation. The teachers judged the summaries to be generally clear, coherent, and usable in class. Readability and grade-level vocabulary received the highest scores. Qualitative comments also noted occasional vocabulary that may be difficult for lower-grade students, incomplete preservation of some source details, and insufficient emphasis on the lesson or educational message.

Table. 7. Summary of Teacher Evaluation Results Using a Likert Scale

Criterion	**Evaluation Question**	**Scale (1-5)**
Content Fidelity	Does the summary accurately preserve the information from the source text without introducing factual or semantic distortions?	4
Key Information Coverage	Are the main ideas and important information adequately retained in the summary?	4
Readability	Is the content readable and easy to understand for primary school students?	5
Grade-Level Vocabulary	Is the vocabulary appropriate for the specified grade level?	5
Coherence	Are the sentences well connected and organized into a coherent summary?	4
Lesson/Message Preservation	Is the main lesson, message, or educational meaning of the source text preserved and sufficiently emphasized?	3
Overall Educational Suitability	Overall, is the summary suitable for supporting classroom learning?	4

4) Discussion and Limitations

The main empirical result is that target grade and Easy/Medium/Hard mode do not collapse into a single length control. As shown in Fig. 3, later grades mainly increase summary length and lexicon, whereas mean sentence length stays in a narrow band. In contrast, Fig. 4 shows that Easy/Medium/Hard mode also raises sentence-level load. Taken together with the high teacher scores on readability and grade vocabulary, this pattern suggests that dual-level constraints can be attached to extractive and abstractive generation and still yield wording that is usable in Vietnamese primary classrooms. VAS remaining competitive with Qwen2.5-3B and Llama-3.2-3B on overlap is consistent with that use; however, that ranking is secondary to the grade- and mode-wise trends. Nevertheless, several limitations remain. The teacher panel is small (N=5) and is not split by grade or by mode, the cost of extra refinement cycles is unreported, and lesson/message preservation is weaker than wording quality. These gaps do not remove the observed trends. They do indicate, however, that vocabulary control, coverage of source details, and pedagogical message preservation still need refinement, together with per-grade and per-mode human scores, before the same interface is fielded as a classroom tool.

V. Conclusion

This paper specified dual-level difficulty control for Vietnamese text summarization in primary education. A target grade (1–5) is encoded by cumulative dictionaries; an Easy, Medium, or Hard mode inside that grade is obtained from a readability score. The controller is shared by planning, extractive and abstractive generation, and revision in the MAS-ViSum multi-agent pipeline. On an educational corpus of textbook and children's texts, the two coordinates produce distinguishable readability profiles. Extractive and abstractive drafts share the same grade–mode checks, and overlap scores for the abstractive branch remain competitive with mid-size LLM baselines. Teachers rate the wording as usable in class. The corpus and the grade-stratified dictionaries can be reused in later Vietnamese educational NLP studies. Future work includes a lesson-aware objective, stronger syntactic control by grade, and a per-grade, per-mode evaluation.

References

[1] M. R. Ramirez-Avila and J. P. Barreiro, "The Effect of Summarizing Narrative Texts to Improve Reading Comprehension," *Journal of Foreign Language Teaching and Learning,* vol. 6, no. 2, p. 4–110, 2021.

[2] A. Avivah, F. Hilmiyati and K. Khaeroni, "The Utilization of Text Summary Techniques to Improve Students' Reading Comprehension Skills," *Journal of Integrated Elementary Education,* vol. 2, no. 2, p. 95–104, 2022.

[3] M. Luo, B. Xue and B. Niu, "A comprehensive survey for automatic text summarization: Techniques, approaches and perspectives," *Neurocomputing,* vol. 603, 2024.

[4] Supriyono, A. P. Wibawa, Suyono and F. Kurniawan, "A survey of text summarization: Techniques, evaluation and challenges," *Natural Language Processing Journal,* vol. 7, 2024.

[5] H. N. Long and H. T. Phú, "Xây dựng ứng dụng AI hỗ trợ tóm tắt bài đọc tiếng Việt cho học sinh tiểu học," *Hong Bang International University Journal of Science,* vol. 36, p. 173–182, 2025.

[6] H. N. Long, N. M. Vũ, P. C. Khanh, L. H. Nghiêm, B. H. C. Cường, and H. T. Phú, "Xây dựng hệ thống đa tác tử hỗ trợ tóm tắt bài đọc tiếng việt bậc tiểu học", *Hong Bang International University Journal of Science*, Special Issue 1, p. 450–460, May 2026.

[7] H. Y. Durak, F. Egin and A. Onan, "Enhancing text summarization with AI: a multi-agent system and human comparison in educational contexts," *Thinking Skills and Creativity,* vol. 59, 2026.

[8] Y. Chen, B. Wen and F. Zulkernine, "A Multiagent Summarization and Auto-Evaluation Framework for Medical Text: Development and Evaluation Study," *MIR AI,* vol. 4, 2025.

[9] D. Q. Nguyen and A. T. Nguyen, "PhoBERT: Pre-trained language models for Vietnamese," *Findings of the Association for Computational Linguistics: EMNLP 2020*, p. 1037–1042, 2020.

[10] L. Phan, H. Tran, H. Nguyen and T. H. Trinh, "ViT5: Pretrained Text-to-Text Transformer for Vietnamese Language Generation," *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Student Research Workshop*, p. 136–142, 2022.

[11] Q.-A. Nguyen, D.-C. Can, H.-Q. Le and M.-V. Tran, "VLSP 2022 Abmusu Task Dataset: A Resource for Vietnamese Abstractive Multi-Document Summarization," *International Journal of Asian Language Processing*, vol. 32, no. 2, 2022.